

Mohamed Mamdouh

Full-Stack .NET & Angular Developer

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SUMMARY

Full-Stack .NET & Angular Developer experienced in building production web applications with ASP.NET Core, Angular, SQL. Strong in API design, secure authentication, Clean Architecture, testing, and CI/CD pipelines.

EDUCATION

Bachelor of Engineering in Computer and Systems Engineering,
Zagazig University — 2026.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, T-SQL, HTML, CSS.

Backend: ASP.NET Core Web API, Entity Framework Core, LINQ, ADO.NET, ASP.NET Identity, JWT, Refresh Tokens, HTTP-only Cookies, Role-Based Authorization (RBAC), Dependency Injection, FluentValidation, MediatR (CQRS), SignalR, Background Services, Caching, Middleware, Logging.

Frontend: Angular, RxJS, Signals, Lazy Loading, Routing, Reactive Forms, HTTP Interceptors, Route Guards, Dependency Injection, Tailwind CSS, Bootstrap.

Databases & Storage: SQL Server, PostgreSQL, MongoDB, Cloudflare R2, Database Design, Query Optimization.

Testing & DevOps: xUnit, Testcontainers, Docker, GitHub Actions (CI/CD), Postman, Swagger, Git, GitHub.

Architecture & Concepts: SOLID Principles, Design Patterns, Clean Code, Clean Architecture, 3-Tier Architecture, Domain-Driven Design fundamentals, Transactional Outbox Pattern, RESTful API Design, SDLC.

SELECTED PROJECTS

Amanah – Lost & Found Platform

[GitHub](#) [Live Production](#)

Full-stack lost-and-found platform that connects people who lost items with finders through moderated listings, structured ownership claims, and in-app chat.

Tech Stack: .NET, Angular, PostgreSQL, EF Core, FluentValidation, JWT Bearer, SignalR, Cloudflare R2, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Implemented API reliability with centralized exception handling, correlation IDs, validation, rate limiting, HybridCache, structured logging, and dependency health checks.
- Implemented phone/email OTP registration, password authentication, short-lived JWT access tokens, and HTTP-only refresh cookies for session persistence.
- Built responsive Arabic RTL interfaces with reactive forms, EXIF metadata stripping, and client-side WebP image processing before storage uploads.
- Integrated SignalR chat with REST fallback and a transactional outbox for reliable SMS and email dispatch.
- Implemented background workers for listing expiration, claim timeouts, storage cleanup, and scheduled maintenance.
- Built integration tests with xUnit and Testcontainers using isolated database instances, and integrated them into GitHub Actions CI pipelines.

Shora – Consulting & Booking Platform

[GitHub](#) [Live Production](#)

Full-stack booking platform for one-to-one consulting sessions with availability management, manual payment verification, client bookings, and admin operations.

Tech Stack: .NET, Angular, PostgreSQL, EF Core, ASP.NET Identity, JWT Bearer, FluentValidation, Cloudflare R2, Brevo, Docker, Render, Supabase, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Prevented double-booking with PostgreSQL row locks, transactions, and partial unique indexes, and modeled booking/payment lifecycles with state machines and audit trails.
- Built manual payment verification for Vodafone Cash/InstaPay with receipt validation, SHA-256 duplicate detection, and secure presigned URLs.
- Designed a scheduling system with recurring availability, blocked dates, booking-hold limits, payment deadlines, and cancellation rules.
- Built a role-based Angular standalone SPA using signals, lazy routes, guards, typed API contracts, and RTL, deployed with Docker, Supabase PostgreSQL, and Cloudflare R2.
- Added job monitoring, admin alerts, API error codes, and earnings reporting for gross, refunded, and net revenue.

MechanicShop – Workshop Operations API

[GitHub](#)

Backend REST API for a mechanic shop covering work orders, day scheduling, labor assignment, customer and vehicle records, and invoicing.

Tech Stack: ASP.NET Core, SQL Server, EF Core, MediatR, JWT, ASP.NET Identity, FluentValidation, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Built a Clean Architecture backend with CQRS/MediatR, Result pattern, FluentValidation, and pipeline behaviors for validation, logging, transactions, and caching.
- Implemented secure JWT authentication with rotating refresh tokens, per-device sessions, reuse detection, RBAC, and custom policies.
- Modeled work orders, day scheduling, labor assignment, and invoicing with optimistic concurrency, with unit/API integration tests with Testcontainers and separate GitHub Actions CI jobs.

DVLD – Driver & Vehicle License Department System

[GitHub](#)

Desktop application for managing driver licenses, license applications, driving tests, and vehicle-related records.

Tech Stack: C#, .NET Framework, Windows Forms, ADO.NET, SQL Server.

Key Highlights:

- Architected a 3-tier desktop solution using C# and WinForms, enforcing clean separation between UI, domain logic, and data access layers.
- Connected to SQL Server via ADO.NET and parameterized stored procedures to execute secure entity operations and rule-driven validation queries.
- Implemented core licensing workflows, including a 3-stage exam system (scheduling, grading, retakes).